False Confidence in Generative Samplers: Exact Physical References Expose Hidden Errors and Diagnostic Blindness

Karthik Gangadharaiah
Independent researcher (KoWerk), Amsterdam, The Netherlands
ORCID: 0009-0009-1713-8624

Abstract

Learned samplers are increasingly proposed for equilibrium sampling and free energies in chemistry, and they are usually judged by self-normalized diagnostics such as the effective sample size (ESS). We show with an elementary proposition that these diagnostics are structurally invariant to the Boltzmann mass α of unvisited basins. Conditioned on not proposing a basin, the importance weights follow the law of the truncated system: the ESS stays high while the free energy shifts by $-k_BT\log(1-\alpha)$ and the nominal delta-method interval contracts around the biased value, so coverage falls toward zero while the basin stays unvisited. In a toy double well where one basin holds 49.9% of the mass, a proposal putting mass 10^{-6} on that basin achieves a median ESS of 98% but covers the exact $\log Z$ in 0% of 400 runs. On exactly enumerated Sherrington-Kirkpatrick instances ($n = 24$), nominal 95% intervals from masked diffusion and a GFlowNet cover the truth in only 8% and 47% of runs at $\beta = 2$, while classical estimators and an autoregressive network (VAN-MADE) stay near 95%. A reverse check that scores the proposal on draws from an independent reference sampler flags the dropped basin where the forward Jensen gap is blind. A RealNVP flow trained by reverse KL on an analytic surrogate of the alanine dipeptide Ramachandran surface drops the $C_{7\mathrm{ax}}$ basin (8.1% of the surrogate's mass) and reports a median ESS of 79% with 10% coverage; the reverse check gives $G_{\mathrm{rev}} = 11.7$ nats. Flows on model landscapes in up to 414 dimensions, including the real LJ_{38} potential, behave the same way, although one control with the correct support also miscovers, so a large reverse gap is not specific to a missing basin. Finally, all-atom simulations of chignolin and Trp-cage and a Langevin sweep of LJ_{38} size the effect and show its single-trajectory analogue: an unbiased run reports nothing about basins it has not reached. Omitting the Trp-cage native basin, 50.3% of our trajectory, would shift the free energy by 0.417 kcal/mol.

1 Introduction

Normalizing flows, diffusion models and generative flow networks (GFlowNets) are proposed as samplers of Boltzmann distributions $p(x) \propto e^{-\beta U(x)}$ for molecules and materials (Noé et al., 2019; Hoogeboom et al., 2022; Satorras et al., 2021; Lipman et al., 2023). The partition function Z of such systems is unknown, so a learned sampler is usually judged by statistics of its own importance weights: the effective sample size (ESS), the Pareto $\hat{k}$, and delta-method intervals for the free energy.

These statistics share a critical failure mode: they are computed exclusively from configurations the sampler proposed, so a basin it never proposes leaves no trace in them. Section 3 formalizes this diagnostic blindness: conditioned on never visiting a basin of Boltzmann mass α, the importance weights follow the law of the problem with that basin deleted. As a result, the free-energy estimate is shifted by $-\beta^{-1}\log(1-\alpha)$ while the ESS remains high and nominal confidence intervals shrink at rate $O(M^{-1/2})$ around the wrong value, so coverage falls toward 0% for as long as the basin stays unvisited (Figure 1). While support coverage is an elementary requirement of importance sampling, modern evaluation culture has overlooked a critical

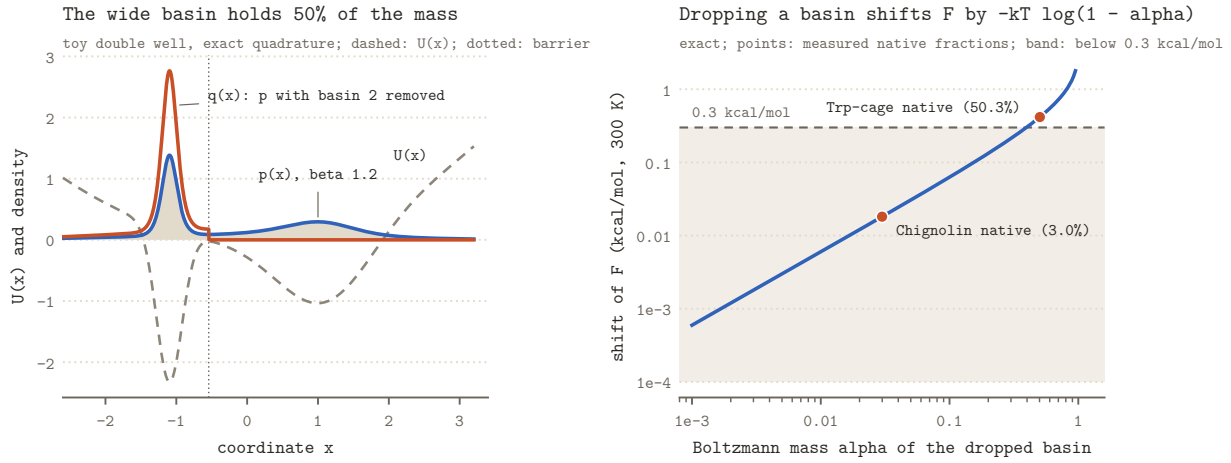


Figure 1: Left: the toy double well of Section 6 at $\beta = 1.2$. The narrow basin is the energy minimum, but the wide basin holds 49.9% of the Boltzmann mass (exact quadrature). Shaded: p. q is p with the wide basin removed. Right: shift of the free energy at 300 K when a basin of mass α is never proposed, $-k_B T \log(1-\alpha)$ (exact). Points mark native-basin fractions; shaded: shifts below 0.3 kcal/mol.

consequence: self-reported diagnostics falsely reward mode collapse with high confidence, reporting near-optimal ESS and contracting intervals precisely when macroscopic states are omitted.

To determine how this pathology manifests across physical systems, we organize our evaluation into three tiers:

1. **Exact empirical audit on disordered systems (Section 5):** Spin glasses feature frustrated, multi-basin landscapes where ground-truth partition functions can be enumerated exactly up to $n = 28$ spins. On these references, masked diffusion and GFlowNets exhibit severe calibration collapse, covering exact values in only 8% and 47% of runs at $\beta = 2$. In contrast, classical thermal estimators and normalized autoregressive networks (VAN-MADE) remain calibrated near 95%.
2. **Continuous neural molecular sampler (Section 6):** On an analytic surrogate of the alanine dipeptide Ramachandran surface, a RealNVP flow trained by reverse KL drops the $C_{7\mathrm{ax}}$ basin (8.1% mass), yet reports a median ESS of 79% while nominal interval coverage collapses to 10%. An independent reverse check immediately flags the omission ($G_{\mathrm{rev}} = 11.7$ nats).
3. **Macromolecular landscape sizing (Section 7):** We size the bound in kcal/mol with all-atom simulations of chignolin and Trp-cage and a Langevin sweep of the LJ_{38} cluster. Omitting the Trp-cage native basin would shift F by 0.417 kcal/mol at the fraction measured in our run; single trajectories also show the blind spot themselves, since they report nothing about basins they have not reached.

Contributions. (i) A formal proposition proving that self-normalized diagnostics and confidence intervals are blind to unvisited Boltzmann mass (Section 3). (ii) A combinatorial benchmark comparing learned versus classical samplers against exact ground truth (Section 5). (iii) A reverse check that flags dropped basins in an exact toy and in flows on model landscapes, with its limits (Section 6). (iv) Molecular simulations that size the shift in kcal/mol and show the single-trajectory analogue (Section 7).

2 Related work

Neural samplers. Variational autoregressive networks (Wu et al., 2019) and Boltzmann generators (Noé et al., 2019) introduced learned samplers for physical systems. Later work uses equivariant diffusion (Hoogeboom et al., 2022; Satorras et al., 2021), flow matching (Lipman et al., 2023; Albergo & Vanden-Eijnden,

2023; Bose et al., 2023), discrete diffusion (Austin et al., 2021; Zhu et al., 2025; Du et al., 2026), annealed transport (Holderrieth et al., 2025) and GFlowNets (Bengio et al., 2021; Malkin et al., 2022).

Free-energy estimators. Bennett's acceptance ratio (Bennett, 1976) and its multistate form (Shirts & Chodera, 2008) combine forward and reverse samples, and the Jarzynski and Crooks relations (Jarzynski, 1997; Crooks, 1999) connect nonequilibrium work to free energies. The reverse check of Section 6 uses the same two-sided idea. Xu (2026) proposes a ground-truth-free check for mode collapse, and Fu et al. (2023) shows that force accuracy does not guarantee correct simulations for learned force fields.

Spin glasses. The Sherrington-Kirkpatrick (SK) model (Sherrington & Kirkpatrick, 1975) has a frustrated landscape whose free energy in the limit $n \to \infty$ is given by the Parisi formula (Parisi, 1980), proven by Guerra (2003) and Talagrand (2006). For small n the partition function is computed exactly by enumeration.

3 Unvisited support

Let $(\Omega, \mathcal{F}, \mu)$ be a configuration space endowed with a dominating reference measure μ (Lebesgue measure for continuous coordinates, counting measure for discrete spins). Let $U : \Omega \to \mathbb{R}$ be a measurable potential energy and $p(x) = e^{-\beta U(x)}/Z$ the Boltzmann probability density with respect to μ, with partition function $Z = \int_\Omega e^{-\beta U(x)}\, d\mu(x)$ and free energy $F = -\beta^{-1} \log Z$. A learned proposal density q_θ (with $q_\theta(x) > 0$ for μ-almost all x where $p(x) > 0$) is trained to approximate p, for example by minimizing $\mathrm{KL}(q_\theta \,\|\, p)$ (Noé et al., 2019). Given i.i.d. draws $x_1, \ldots, x_M \sim q_\theta$, the importance-sampling estimate is

$$w_k = \frac{e^{-\beta U(x_k)}}{q_\theta(x_k)}, \qquad \hat{Z}_M = \frac{1}{M} \sum_{k=1}^{M} w_k, \qquad \hat{F}_M = -\beta^{-1} \log \hat{Z}_M, \tag{1}$$

its reliability is judged by the self-normalized effective sample size

$$\mathrm{ESS}_M = \frac{\left(\sum_k w_k\right)^2}{\sum_k w_k^2}, \tag{2}$$

and a nominal 95% interval is $\hat{F}_M \pm 1.96\, \beta^{-1} \hat{\sigma}_w / (\sqrt{M}\, \hat{Z}_M)$, where $\hat{\sigma}_w$ is the sample standard deviation of the weights.

Proposition 1 (Diagnostic blindness under unvisited support)**.** *Let $A, B \in \mathcal{F}$ form a measurable partition of Ω ($A \cap B = \emptyset$, $A \cup B = \Omega$) with $p(B) = \alpha \in (0, 1)$ and $q_\theta(B) = \epsilon \in [0, 1)$. Let $\mathcal{E}$ be the event that no draw lands in B (all M draws land in A). Then*

1. *$\mathbb{P}(\mathcal{E}) = (1 - \epsilon)^M$, which tends to 1 as $\epsilon \to 0$ for fixed M;*
2. *conditioned on $\mathcal{E}$, the draws are conditionally i.i.d. with density $q_A(x) = q_\theta(x)\mathbf{1}_A(x)/(1 - \epsilon)$;*
3. *conditioned on $\mathcal{E}$, $\mathbb{E}[\hat{Z}_M \mid \mathcal{E}] = (1 - \alpha)Z/(1 - \epsilon)$, and by continuous mapping $\hat{F}_M \xrightarrow{p} F - \beta^{-1} \log \frac{1-\alpha}{1-\epsilon}$ as $M \to \infty$, which approaches $F - \beta^{-1} \log(1 - \alpha)$ when $\epsilon \ll \alpha$;*
4. *conditioned on $\mathcal{E}$, the joint law of the normalized weights $\tilde{w}_k = w_k / \sum_j w_j$, and hence of ESS_M and of the relative interval width $\hat{\sigma}_w / \hat{Z}_M$, depends on p and q_θ solely through their conditional restrictions on A, and is strictly invariant to α and to the potential landscape on B, provided U on A is held fixed.*

Proof. Because the draws x_k are i.i.d., $\mathbb{P}(\mathcal{E}) = \prod_{k=1}^{M} \mathbb{P}(x_k \in A) = (1 - \epsilon)^M$. Conditioned on the product event $\mathcal{E}$, each draw has conditional density $q_\theta(x \mid x \in A) = q_A(x)$ independently. For $x \in A$,

$$\mathbb{E}_{q_A}[w] = \int_A \frac{e^{-\beta U(x)}}{q_\theta(x)} \frac{q_\theta(x)}{1 - \epsilon}\, d\mu(x) = \frac{Z_A}{1 - \epsilon}, \qquad Z_A = \int_A e^{-\beta U(x)} d\mu(x) = (1 - \alpha)Z. \tag{3}$$

By the weak law of large numbers conditioned on $\mathcal{E}$, $\hat{Z}_M \xrightarrow{p} (1-\alpha)Z/(1-\epsilon)$. Since $g(z) = -\beta^{-1}\log z$ is continuous on $(0,\infty)$, the continuous mapping theorem yields $\hat{F}_M \xrightarrow{p} F - \beta^{-1}\log\frac{1-\alpha}{1-\epsilon}$, proving item 3. For item 4, write $e^{-\beta U(x)} = Z_A\, p(x \mid A)$ for $x \in A$. Then $w_k = c\, p(x_k \mid A)/q_A(x_k)$ where $c = Z_A/(1-\epsilon)$ is a positive scaling constant. The normalized weights $\tilde{w}_k = w_k/\sum_j w_j = \frac{p(x_k|A)/q_A(x_k)}{\sum_j p(x_j|A)/q_A(x_j)}$ cancel c identically, so their joint distribution depends on p and q_θ solely through their restrictions to A. Changing α or altering U on B while preserving $U|_A$ only shifts c, leaving the joint law of the normalized weights, ESS_M, and $\hat{\sigma}_w/\hat{Z}_M$ strictly unchanged. □

Diagnostic consequence: coverage collapse. The practical danger is not merely that importance sampling requires support coverage: it is the structural blindness of self-reported metrics. By item 4, if a model places negligible mass $\epsilon \ll 1/M$ on a basin, the event $\mathcal{E}$ occurs with high probability ($\approx 99.9\%$ for $\epsilon = 10^{-6}$, $M = 10^3$). Conditioned on $\mathcal{E}$, the sample variance $\hat{\sigma}_w^2$ and tail diagnostics such as Pareto $\hat{k}$ (Vehtari et al., 2024) reflect only the fit within A. In any regime where $\epsilon \ll 1/M$ as sample size M grows, the nominal delta-method confidence interval contracts at rate $O(M^{-1/2})$ around the biased target $F - \beta^{-1}\log(1-\alpha)$, while ESS_M/M can approach 100% and $\hat{k}$ can remain falsely reassuring (< 0.5). Consequently, the empirical coverage of the nominal 95% interval collapses asymptotically to 0%.

In continuous state spaces, A and B are Borel measurable sets, and $q_A(x) = q_\theta(x)/\int_A q_\theta$ is well-defined whenever $q_\theta(x) > 0$ on A. While continuous neural proposals typically have non-zero support everywhere due to diffuse Gaussian tails, this pathology remains operant whenever a physical basin is separated by high free-energy barriers or entropic bottlenecks. If the proposal assigns probability $\epsilon \ll 1/M$ to that basin, no finite sample will visit it, leaving the empirical weights truncated and the diagnostics blind.

The right panel of Figure 1 illustrates the scale of this bias: at 300 K a dropped basin must hold about 50.3% of the mass (as Trp-cage does in our trajectory) to shift F by 0.417 kcal/mol, exceeding common chemical accuracy targets, while a basin holding 3.0% (chignolin in our trajectory) shifts F by only 0.018 kcal/mol. Related failure modes for mode-collapsed discrete samplers were observed by Xu (2026).

4 Why spin glasses are a fair test

A natural objection is that discrete Ising spins on a complete graph are physically distinct from continuous molecular conformations in three dimensions. The mathematical statement of Proposition 1 is coordinate-free: it requires only that the target distribution possess multiple metastable basins separated by barriers, and that the proposal fails to populate at least one of them. What an empirical benchmark requires is a system with many such frustrated basins where the true partition function is known exactly, without relying on approximate reference simulations.

The Sherrington-Kirkpatrick (SK) model, $H_n(\sigma) = n^{-1/2}\sum_{i<j} g_{ij}\sigma_i\sigma_j$ with $g_{ij} \sim \mathcal{N}(0,1)$, provides exactly this setting. Below the critical temperature ($\beta > 1$), its free-energy landscape fractures into a hierarchical ultrametric tree of pure states with extensive Hamming barriers (Parisi, 1980; Talagrand, 2006). For $n \leq 28$ spins ($2^{28} \approx 2.68 \times 10^8$ configurations), the complete state space can be enumerated exactly (a Gray-code enumerator in Rust in the companion benchmark), yielding exact partition functions and Boltzmann weights.

We do not claim that mean-field spin glasses replicate the Euclidean geometry, bond constraints, or continuous transition paths of atomistic proteins. The role of the SK model is a rigorous falsification test: if an architecture or diagnostic cannot reliably detect the omission of macroscopic states in an exactly-solvable system with known truth, it provides no mathematical or empirical justification for confidence on continuous molecular landscapes where ground truth is unknown.

5 Exact audit on spin glasses

We evaluate the main test split of the SpinGlassBench benchmark at $n = 24$: Gaussian SK instances, seven inverse temperatures $\beta \in \{0.3, 0.6, 0.9, 1, 1.2, 1.5, 2\}$, an equal budget of 10^5 energy evaluations per estimate across all methods, exact $\log Z$ from Gray-code enumeration, and 5 seeds per instance. Classical

Table 1: SK, $n = 24$, test split, 10^5 energy evaluations per estimate, 5 seeds per instance. Error: mean $|\log \hat{Z} - \log Z|/n$. Coverage: share of nominal 95% intervals containing exact $\log Z$ (Clopper-Pearson 95% binomial CIs in text); n/a: method reports no interval. Seconds: mean wall-clock per estimate at $\beta = 1.5$, one thread; learned methods include training.

Method	$\lvert\log \hat{Z} - \log Z\rvert/n$		Coverage (%)		Seconds
	$\beta = 0.6$	$\beta = 1.5$	$\beta = 0.6$	$\beta = 1.5$	per estimate
SA-TI	2.4×10^{-4}	7.1×10^{-4}	n/a	n/a	0.06
PT-TI	2.8×10^{-4}	8.7×10^{-4}	n/a	n/a	0.11
AIS	2.7×10^{-4}	8.9×10^{-4}	96	94	0.07
SMC	2.7×10^{-4}	8.8×10^{-4}	96	95	0.07
PA	2.7×10^{-4}	7.8×10^{-4}	96	96	0.08
VAN-MADE	2.7×10^{-5}	8.6×10^{-5}	95	96	2.49
Masked diffusion	1.6×10^{-4}	1.3×10^{-2}	94	60	6.96
GFlowNet	1.5×10^{-3}	2.4×10^{-2}	94	71	30.60

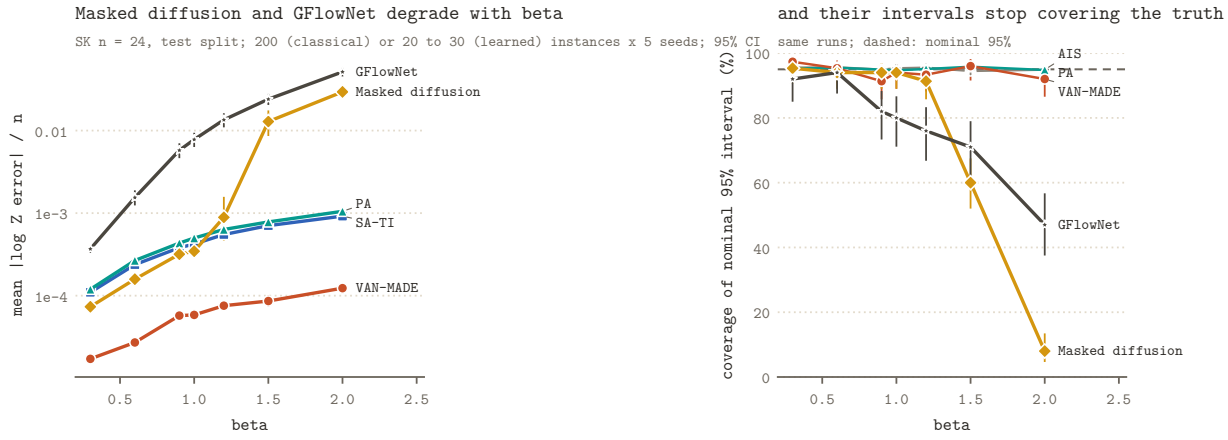


Figure 2: Left: mean absolute error of $\log Z$ per spin against β. Right: coverage of nominal 95% intervals. SK $n = 24$, test split, 5 seeds per instance; error bars are 95% intervals over instances. Dashed: nominal 95%.

estimators were evaluated across 200 instances (1,000 runs total); computationally intensive learned methods were trained and evaluated across 20 to 30 instances (100 to 150 runs total). Table 1 and Figure 2 report summary metrics from this benchmark.

Accuracy and architectural divergence. At $\beta = 1.5$, error per spin is 7.1×10^{-4} for simulated annealing with thermodynamic integration (SA-TI) and 7.8×10^{-4} for population annealing (PA). Crucially, learned samplers show sharp architectural divergence:

- **Normalized autoregressive networks (VAN-MADE):** Achieve the highest accuracy per energy evaluation (8.6×10^{-5}, roughly 8 times lower error than SA-TI) and maintain calibrated coverage across all temperatures: 95% at $\beta = 0.6$ (143/150 runs; 95% Clopper-Pearson interval $[90.6\%, 98.1\%]$), 96% at $\beta = 1.5$ ($[91.5\%, 98.5\%]$), and 92% at $\beta = 2$ ($[86.4\%, 95.8\%]$), close to PA at $\beta = 2$ (95%, 948/1000 runs, $[93.2\%, 96.1\%]$). A possible reason, which we have not tested, is that an autoregressive model's exactly normalized probabilities make the reverse-KL gradient sensitive to missing mass. However, this relies on a 1D sequence factorization ($p(\sigma) = \prod_i p(\sigma_i \mid \sigma_{<i})$). In continuous 3D molecular coordinates, autoregressive factorizations over all atom positions are geometrically constrained and break Euclidean equivariance, which is one reason diffusion models and continuous flows are more common in chemistry.

- **Iterative generative samplers (Masked Diffusion and GFlowNets):** Are one to two orders of magnitude less accurate (1.3×10^{-2} and 2.4×10^{-2}) and undergo catastrophic calibration collapse in the spin-glass phase ($\beta \geq 1.5$). Masked diffusion coverage falls from 94% ([88.9%, 97.2%]) at $\beta = 0.6$ to 60% ([51.7%, 67.9%]) at $\beta = 1.5$ and 8% (12/150 runs, [4.2%, 13.6%]) at $\beta = 2$. GFlowNet coverage drops to 71% ([61.1%, 79.6%]) and 47% (47/100 runs, [36.9%, 57.2%]).

On one test instance, illustrated in the companion paper, the diffusion sampler keeps one side of the overlap distribution and the GFlowNet concentrates on a few overlaps. By Proposition 1, this omitted mass does not widen nominal intervals, producing false confidence.

Wall-clock trade-offs. Table 1 reports single-thread CPU execution time, with learned methods including training. Under this metric, SA-TI finishes in 0.06 s and PA in 0.08 s, whereas VAN-MADE requires 2.49 s and diffusion requires 6.96 s. While neural samplers can amortize training costs when millions of samples are required, on single-target free-energy evaluations classical annealing remains substantially faster.

External code audits. The companion SpinGlassBench study used the same exact references to score released checkpoints of published samplers; we summarize its results. In MDNS (Zhu et al., 2025), the released free-energy estimator rounds one per-step term in bfloat16, which biases $\log Z$ upward by up to 4.6×10^{-4} per site; a one-line change removes the bias. MetaDNS (Du et al., 2026), which shares this code, is unbiased at its float16 default. In LEAPS (Holderrieth et al., 2025), free energies at the released time grids are low by 2.2 to 3.2×10^{-3} per site, mostly from time discretization. The ESS did not reveal these errors: for MDNS the biased and corrected estimators have nearly the same ESS, and for LEAPS the ESS fraction is 0.82 on Ising (biased) and 0.31 on Potts, where it is no higher at the finer grids that reduce the bias.

6 A reverse check

No statistic computed from draws of q_θ alone can detect an unvisited basin (Proposition 1). Breaking this blindness requires configurations that q_θ did not propose. Suppose an independent reference sampler, such as parallel tempering or annealed importance sampling, provides $y_1, \ldots, y_K$ approximately distributed according to p. Writing $\ell(x) = -\beta U(x) - \log q_\theta(x)$ for the unnormalized log importance weight, two diagnostic gaps arise from Jensen's inequality:

$$G_{\mathrm{fwd}} = \log \hat{Z}_M - \frac{1}{M} \sum_{k=1}^{M} \ell(x_k), \qquad G_{\mathrm{rev}} = \frac{1}{K} \sum_{j=1}^{K} \ell(y_j) - \log \hat{Z}_M. \tag{4}$$

By Jensen's inequality applied to the concave logarithm, $\log(\frac{1}{M} \sum_k e^{\ell(x_k)}) \geq \frac{1}{M} \sum_k \ell(x_k)$, so $G_{\mathrm{fwd}} \geq 0$ always. When q_θ covers the support of p, G_{fwd} serves as an empirical estimator of $\mathrm{KL}(q_\theta \,\|\, p)$. However, when a basin is omitted, G_{fwd} reflects only the fit within the visited region A, approaching $\mathrm{KL}(q_A \,\|\, p_A) \approx 0$ if q_θ fits A well.

Conversely, because $\mathbb{E}_p[\ell] = \log Z + \mathrm{KL}(p \,\|\, q_\theta)$, the reverse gap satisfies

$$\mathbb{E}_{p,q_\theta}[G_{\mathrm{rev}}] = \mathrm{KL}(p \,\|\, q_\theta) + (\log Z - \mathbb{E}_{q_\theta}[\log \hat{Z}_M]). \tag{5}$$

Both terms expand substantially when a basin of mass α is dropped: by data processing, $\mathrm{KL}(p \,\|\, q_\theta) \geq \alpha \log(\alpha/\epsilon) + (1-\alpha) \log \frac{1-\alpha}{1-\epsilon}$ with $\epsilon = q_\theta(B) \to 0$, and $\log Z - \log \hat{Z}_M \approx -\log(1-\alpha)$. The forward gap sees neither. For a dropped basin of mass $\alpha = 0.081$ with $\epsilon = 10^{-6}$ (in the alanine surrogate we only know that no draw in 5×10^5 landed in $C_{7\mathrm{ax}}$, so $\epsilon \lesssim 10^{-5}$), these two terms contribute about 0.9 nats (with 1 nat ≈ 0.596 kcal/mol at 300 K), while density mismatch elevates G_{rev} to 11.7 nats. We do not propose a universal threshold: in Table 2 a mixture with the correct support reaches 10.33 nats. This mirrors Bennett's acceptance ratio (Bennett, 1976) adapted for mode coverage.

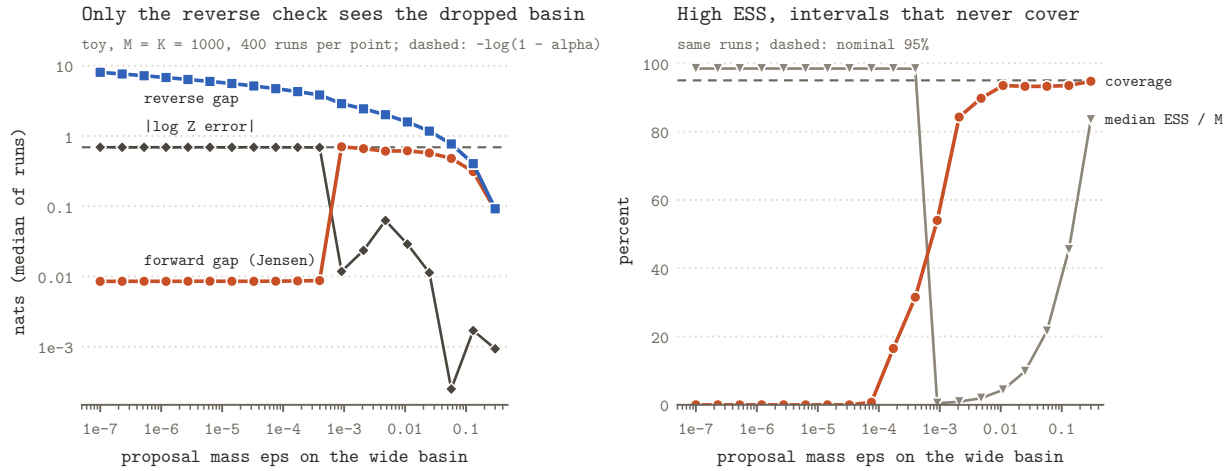


Figure 3: Exact toy, $M = K = 1{,}000$, 400 runs per value of ϵ. Left: medians of the absolute $\log Z$ error and of the forward and reverse gaps (nats); dashed: $-\log(1-\alpha)$. Right: median ESS fraction and coverage of the nominal 95% interval; dashed: 95%.

Operational role: auditing amortized models. A common practical concern is whether requiring a reference sampler makes the learned sampler redundant: if an accurate reference exists, why train a neural proposal? The answer lies in amortization. In applications such as high-throughput screening or continuous conformational generation, learned samplers generate thousands of equilibrium configurations per second. Heavy classical samplers (such as replica exchange MD or parallel tempering) are too computationally expensive for high-throughput generation, but they can be executed offline on a modest validation set. By evaluating the reverse gap G_{rev} on these benchmark targets, practitioners can check whether the amortized network has dropped basins that the reference visits. Furthermore, the reference sampler does not need to be mathematically exact: as long as it visits the basin missed by q_θ, the reverse gap grows with the omission; a large gap can also occur with the correct support (Table 2), so it calls for a closer look rather than proving a dropped basin. Standard multi-chain diagnostics (such as Gelman-Rubin $\hat{R} < 1.05$ or replica-exchange round-trip rates) can be used to ensure the offline reference set has visited the relevant basins before auditing q_θ.

Exact toy validation. The double well of Figure 1 has

$$U(x) = -2.5\, e^{-(x+1.1)^2/(2\cdot 0.18^2)} - 1.2\, e^{-(x-1.1)^2/(2\cdot 0.65^2)} + 0.15\, x^2, \qquad \beta = 1.2. \tag{6}$$

By numerical quadrature, the two basins (divided at the barrier peak, $x = -0.54$) have Boltzmann weights 5.88 and 5.85. The wider basin holds $\alpha = 49.9\%$ of the equilibrium mass despite having a higher local energy minimum, due to entropic width. The proposal mixes tempered densities of the two basins, placing mass $1-\epsilon$ on the narrow well and ϵ on the wide well. We draw $M = 1{,}000$ proposals and $K = 1{,}000$ exact reference draws in each of 400 independent runs per value of ϵ.

For $\epsilon \le 10^{-4}$, almost no run proposes the wide basin (at $\epsilon = 10^{-6}$, 99.8% of runs). In these runs, the median ESS is 98% and the forward Jensen gap is nearly zero (0.009 nats), falsely signaling excellent convergence. Meanwhile, the true $\log Z$ error is 0.69 nats (matching the theoretical shift $-\log(1-\alpha) = 0.69$), and nominal 95% intervals cover the true free energy in 0% of runs (Figure 3). In sharp contrast, the reverse gap detects the omission immediately, reporting 6.8 nats.

A continuous flow on an alanine surrogate. To evaluate Proposition 1 and the reverse check on a continuous model, we trained a 4-layer RealNVP normalizing flow (Noé et al., 2019) via reverse-KL divergence on an analytic two-dimensional surrogate of the alanine dipeptide Ramachandran surface (ϕ, ψ) at 300 K: three Gaussian wells ($C_{7\text{eq}}$, α_R, $C_{7\text{ax}}$) and a barrier at $\phi = 0$. It is not a force field, and its basin masses

are properties of the surrogate. By fine 2D quadrature (300×300 grid), the metastable $C_{7\mathrm{ax}}$ basin ($\phi > 0$) accounts for 8.1% of the equilibrium Boltzmann mass ($F_{\mathrm{exact}} = -3.392\,\mathrm{kcal/mol}$).

Under reverse-KL training, the flow collapses into the deeper $C_{7\mathrm{eq}}$ basin and proposes $< 0.01\%$ of draws in $C_{7\mathrm{ax}}$. Despite completely omitting this state, across 100 independent evaluation runs ($M = 5{,}000$ draws each), the flow reports a median internal ESS of 79% and nominal 95% confidence intervals covering the exact free energy in only 10% of runs ($\pm 0.048\,\mathrm{kcal/mol}$ interval width around a $+0.122\,\mathrm{kcal/mol}$ biased estimate).

The forward Jensen gap reports $G_{\mathrm{fwd}} = 0.028\,\mathrm{nats}$, blind to the missing fold. However, the reverse check evaluated on independent reference draws immediately spikes to $G_{\mathrm{rev}} = 11.7\,\mathrm{nats}$. Most of this gap comes from reference draws in $C_{7\mathrm{ax}}$, where the flow's density is tiny; we did not store the per-basin decomposition.

Higher-dimensional model landscapes. We repeated the audit with flows in 66 to 414 Cartesian dimensions on three landscapes whose $\log Z$ is known (Table 2, Figure 4): two analytic surrogates, for which $\log Z$ is exact, and the real Lennard-Jones potential of LJ_{38} restricted to two basins, for which it is estimated. None is a biomolecular force field, and none tests whether a published molecular sampler drops a basin.

1. **Alanine surrogate in 3D (66D).** The analytic Ramachandran surface above plus harmonic bond, angle and stiff-torsion terms, mapped to $\mathbb{R}^{22\times 3}$ through internal coordinates (1,549,248-parameter coupling flow). The flow trained by reverse KL samples $C_{7\mathrm{ax}}$ in fewer than 0.01% of draws, reports a median ESS of 91%, and its intervals cover in 2.5% of runs; $G_{\mathrm{fwd}} = 0.010$ and $G_{\mathrm{rev}} = 7.65\,\mathrm{nats}$. The control, a mixture of two flows with the exact basin weights, also fails: it covers in 1.5% of runs with error $0.094\,\mathrm{kcal/mol}$ and $G_{\mathrm{rev}} = 10.33\,\mathrm{nats}$. Its error cannot come from a missing basin, since it samples $C_{7\mathrm{ax}}$ at the equilibrium rate; we did not identify its source. In this case the reverse gap does not separate the two proposals. The single flow's error ($0.153\,\mathrm{kcal/mol}$) is also larger than the $0.050\,\mathrm{kcal/mol}$ a dropped basin alone explains.

2. **LJ_{38} two-basin model (114D).** The standard Lennard-Jones potential of the cluster (it reproduces the stored minimum energies, -173.928 and -173.252) at $T = 0.012$. Each basin's $\log Z$ is estimated by MALA started at its minimum and BAR against a harmonic reference (32 chains, $\hat{R} < 1.012$; standard error 0.004). Other minima and permutational isomers are not included, so this is a two-basin approximation of the cluster's partition function. At this temperature the FCC basin holds almost all the mass. A RealNVP flow and an $SE(3)$-invariant distance flow, each confined to the icosahedral basin, report median ESS of 45% and 45% with 0% coverage, while G_{rev} is 58,724 and 65,059 nats. The Pareto $\hat{k}$ of the $SE(3)$ flow's weights is -0.92. Two-flow mixtures with fixed equal weights cover in 97% of runs.

3. **Two-Gaussian 408D model.** Two anisotropic Gaussian wells in the 408 internal Cartesian coordinates of a 138-atom chignolin structure, one at the experimental structure and one displaced in a random direction, with the offset set so that the first well holds 68% of the mass. The energy is not Amber14SB, and the 68% is an input taken from the literature value discussed in Section 7. A flow confined to the second well reports a median ESS of 83% and 0% coverage, with error $0.680\,\mathrm{kcal/mol}$, equal to $-k_B T \log(1 - 0.68)$ as Proposition 1 predicts. G_{rev} is 32,434 nats, against 0.096 nats for the two-flow mixture with the exact weights, which covers in 95% of runs.

The controls are mixtures of two separately trained flows with fixed weights, not single samplers that found both basins; they show what a proposal with the right support reports, not that such a proposal is easy to train. We computed the Pareto $\hat{k}$ from the flows' own weights only for the $SE(3)$ model; the values in the v2 preprint for the other flows were not computed that way and are withdrawn.

7 Molecular simulations

The spin-glass benchmark provides exact reference values on discrete configurations, and Section 6 demonstrated the failure mode on flows over continuous model landscapes. In this section, we translate the theoretical shift of Proposition 1 into chemical units (kcal/mol) across macromolecular folding landscapes, and

Table 2: Flows on landscapes with known $\log Z$: two analytic surrogates (alanine, two Gaussian wells) and the real LJ_{38} potential restricted to two basins. ESS: median over runs ($R = 100$ for the 2D model, 200 otherwise). Cov.: share of nominal 95% intervals containing the exact F. $|\hat{F} - F|$: mean absolute error. G_{rev}: median reverse gap. Mixtures combine two separately trained flows with fixed weights.

Landscape	Proposal	Params per flow	Support	ESS (%)	Cov. (%)	$\lvert\hat{F} - F\rvert$	G_{rev} (nats)
Alanine 2D surrogate	RealNVP, reverse KL	4 layers	C7eq only	79	10.0	0.12 kcal/mol	11.66
Alanine 3D surrogate	NeRF RealNVP	1,549,248	C7eq only	91	2.5	0.15 kcal/mol	7.65
	two-flow mixture	1,549,248	both basins	86	1.5	0.094 kcal/mol	10.33
LJ_{38} two-basin	Eckart RealNVP	1,943,448	ico only	45	0.0	0.65 ϵ	58,724
	two-flow mixture	1,943,448	both basins	24	97.0	0.0002 ϵ	1.05
LJ_{38} two-basin	SE(3) distance flow	2,643,864	ico only	45	0.0	0.65 ϵ	65,059
	two-flow mixture	2,643,864	both basins	23	97.0	0.0002 ϵ	1.04
Two-Gaussian 408D	Eckart RealNVP	10,902,704	basin B only	83	0.0	0.68 kcal/mol	32,434
	two-flow mixture	10,902,704	both basins	83	95.0	0.0031 kcal/mol	0.096

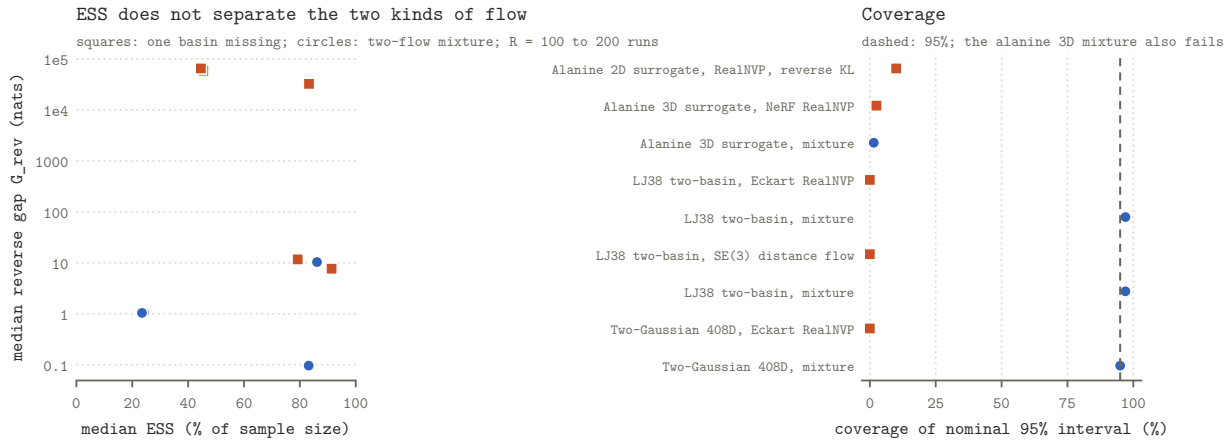


Figure 4: Flows of Table 2. Left: median reverse gap against median ESS; squares: flow missing one basin; circles: two-flow mixture. Right: interval coverage; dashed: 95%.

examine what conventional molecular dynamics trajectories reveal about kinetic trapping versus converged equilibrium thermodynamics.

Simulation parameters. All-atom simulations were executed in OpenMM using the Amber14SB force field with OBC2 implicit solvent at 300 K (Langevin middle integrator, 2 fs timestep, friction coefficient 1 ps^{-1}). Production runs spanned 1.0 μs (5.0×10^8 steps, 100,000 frames) for chignolin and 300 ns (1.5×10^8 steps, 30,000 frames) for Trp-cage, saving configurations every 10 ps. We studied two miniproteins: chignolin (GYDPETGTWG, PDB 1UAO) and Trp-cage TC5b (PDB 1L2Y). For each frame, we computed the Cα RMSD against the experimental structure following optimal superposition, and radius of gyration R_g. A frame is designated native if its Cα RMSD is below 2 Å. Uncertainties on observed fractions are 95% bootstrap intervals over ten equal blocks (100 ns for chignolin, 30 ns for Trp-cage). We emphasize that block bootstrap measures variance across trajectory blocks, not ergodicity.

Chignolin: Kinetic trapping versus equilibrium benchmarks. The continuous 1.0 μs trajectory starts in the native hairpin and leaves it after 1.5 ns (the first stretch of at least 5 ns with no native frame), then returns intermittently, with 11% and 10% native frames in the 700 to 800 and 900 to 1000 ns blocks; the last native frame is at 960 ns (Figure 5). Across the full microsecond (100,000 frames), the native state accounts for only 3.0% of frames (ten 100 ns block fractions: 3.7%, 0.6%, 0%, 0%, 1.9%, 0.4%, 0%, 10.6%, 3.0%, 9.8%), reflecting persistent kinetic trapping in a non-native misfolded globule. If this non-ergodic fraction represented equilibrium, omitting the native state would shift free energy by only 0.018 kcal/mol.

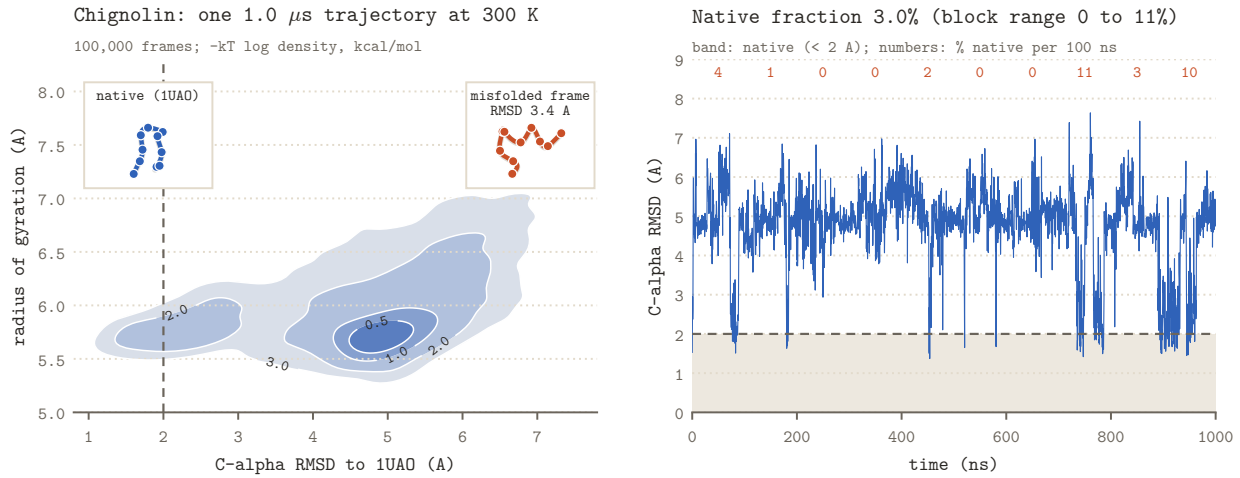


Figure 5: Chignolin (PDB 1UAO), one 1.0 μs trajectory at 300 K, Amber14/OBC2. Left: $-k_BT$ log density in (RMSD, R_g), kcal/mol relative to the maximum density, from a kernel density estimate on 8,000 frames; darker is more populated; dashed: native threshold 2 Å; framed insets: Cα traces of the native structure (1UAO) and of one misfolded frame, with its RMSD to native (illustrations, not data). Right: Cα RMSD against time; shaded: native region; numbers give the percentage of native frames in each 100 ns block.

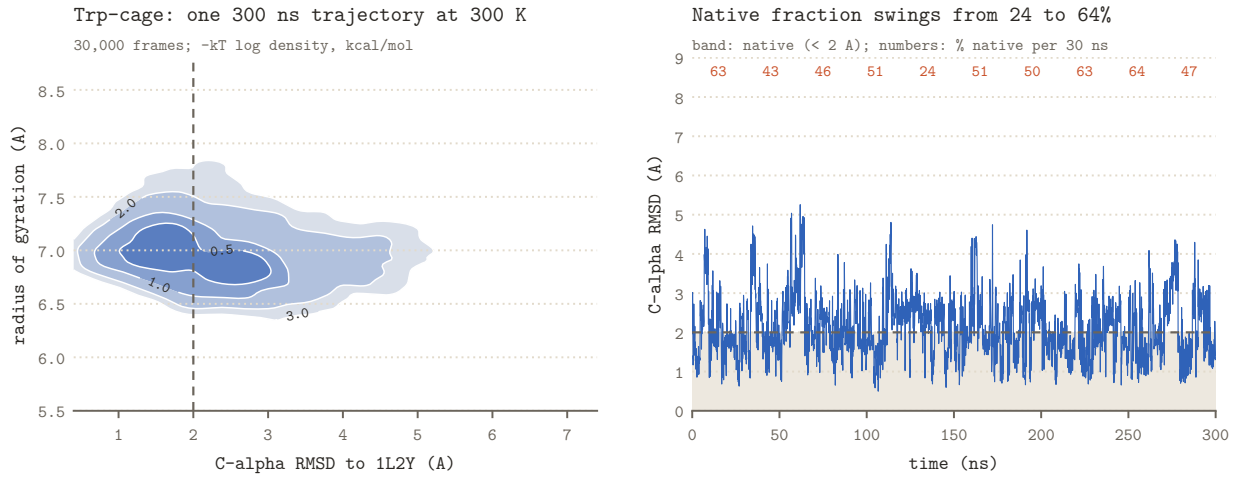


Figure 6: Trp-cage TC5b (PDB 1L2Y), one 300 ns trajectory at 300 K, Amber14/OBC2. Panels as in Figure 5.

Literature estimates of the equilibrium folded fraction are much higher. We use 68% as an illustrative value; it is typed in from our reading of Lindorff-Larsen et al. (2011), and we have not checked it against the source (UNVERIFIED). That study concerns CLN025 (structure in Honda et al., 2008), simulated with CHARMM22* at 340 K (McKiernan et al., 2017), not the GYDPETGTWG peptide simulated here with Amber14SB and implicit solvent at 300 K, so the value is not an equilibrium population for our system. If the native basin held 68% of the mass, omitting it would shift F by 0.68 kcal/mol. The point that survives these caveats is general: a single trajectory can badly underestimate a basin's mass, and the shift from omitting a basin grows quickly with that mass (Figure 1).

Trp-cage: Macroscopic tertiary packing. Trp-cage fluctuates between the native fold and partially unfolded intermediates throughout the 300 ns run (Figure 6; minimum RMSD 0.50 Å). The native basin represents 50.3% of the trajectory (block interval 43.0 to 56.8%), corresponding to a free-energy shift of 0.417 kcal/mol (interval 0.335 to 0.501 kcal/mol). As an illustrative literature value we use a folded fraction

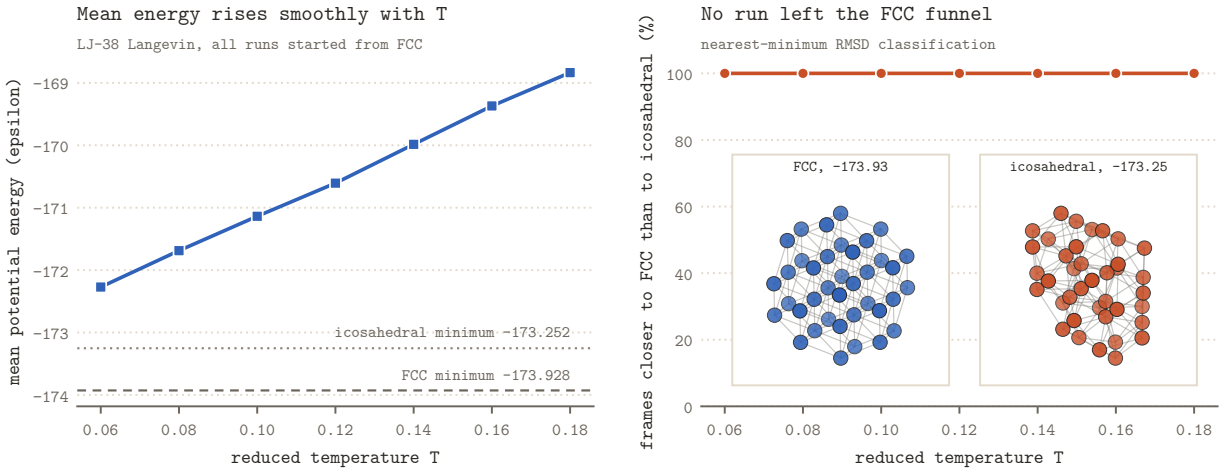


Figure 7: LJ_{38}, Langevin dynamics in reduced units, 7 temperatures from 0.06 to 0.18, every run started from the FCC minimum, 200 stored frames per temperature. Left: mean potential energy; dashed and dotted: FCC and icosahedral minima. Right: share of stored frames whose RMSD is lower to the FCC minimum than to the icosahedral minimum; framed insets: the FCC (global) and icosahedral minima with their energies (illustrations, not data).

of 82%, typed in from our reading of Neidigh et al. (2002) and Lindorff-Larsen et al. (2011) and not checked against the sources (UNVERIFIED; conditions in those studies differ from ours). It would give a shift of 1.02 kcal/mol. With either the measured or the literature fraction, omitting the native basin shifts F by more than 0.3 kcal/mol.

LJ_{38} and double-funnel blindness. The 38-atom Lennard-Jones cluster provides a canonical benchmark for double-funnel landscapes (Doye et al., 1999; Wales et al., 2000). Its potential energy global minimum is a face-centered cubic truncated octahedron (FCC, energy −173.928), while a dense manifold of icosahedral structures resides slightly higher in potential energy (−173.252). The literature reports that the icosahedral funnel becomes the more probable one above a temperature near 0.12 in reduced units (Doye et al., 1999); we have not checked this or the size of its mass against the source (UNVERIFIED).

We performed unbiased Langevin sweeps across 7 temperatures from 0.06 to 0.18, initializing all trajectories in the FCC minimum. As shown in Figure 7, 100% of stored frames remain classified as FCC across all temperatures. In our runs no trajectory escaped the funnel it started in. This provides a clear physical analog to generative mode collapse: just as an unbiased trajectory trapped in the FCC funnel produces internal statistics that betray no existence of the icosahedral basin, a learned proposal confined to one funnel would report a high ESS while missing the other, as the two-basin model of Section 6 shows at low temperature. Accurately resolving both funnels requires enhanced-sampling techniques like parallel tempering, highlighting the necessity of independent reference checks.

8 Limitations

Model landscapes. The flows of Section 6 run on two analytic surrogates (an alanine Ramachandran surrogate and two Gaussian wells in 408 dimensions) and on the real LJ_{38} potential restricted to two basins at low temperature; none is a biomolecular force field. They show that the diagnostics behave as Proposition 1 predicts in high dimension and with an $SE(3)$-invariant architecture; they do not show that any published molecular sampler drops a basin. The controls are mixtures of two flows with fixed weights. In the alanine 3D case the control also miscovers and has a larger reverse gap than the flow that drops the basin, so a large G_{rev} does not by itself show that a basin is missing; we did not identify what causes the control's error and gap. The Pareto $\hat{k}$ was computed from flow weights only for the $SE(3)$ model.

Architectural scope and training objectives. On the exact spin-glass benchmark, our neural evaluation focused on two iterative samplers (masked diffusion and GFlowNets) and one normalized autoregressive model (VAN-MADE), all trained via reverse-KL or variational objectives. These objectives are inherently mode-seeking; while forward-KL or hybrid objectives theoretically encourage mode covering, they require pre-existing equilibrium training data that autonomous samplers seek to avoid. Furthermore, while VAN-MADE's 1D autoregressive factorization penalizes mode omission, this mechanism does not straightforwardly transfer to continuous 3D molecular coordinates, where sequential orderings break SE(3) roto-translational equivariance.

Trajectories, equilibrium baselines, and kinetics. Each system was simulated once (chignolin 1 μs, Trp-cage 300 ns, LJ_{38} short Langevin runs from the FCC minimum). These trajectories are not equilibrated: chignolin leaves the native basin within 1.5 ns and returns only intermittently, and no LJ_{38} run left the FCC funnel. Fractions from them describe the runs, not force-field equilibria. The literature populations we quote for comparison are typed in, unverified against their sources, and for chignolin refer to a different sequence (CLN025). Our analysis concerns static free energies, not kinetics.

Reference dependence, one-sided guarantees, and calibration. The reverse check G_{rev} provides a *one-sided necessary condition* rather than an autonomous ergodicity solver. In expectation G_{rev} exceeds $\alpha \log(\alpha/\epsilon)$ plus the $\log Z$ bias (Section 6), so a large gap signals mode omission or severe density mismatch, and Table 2 shows it does not say which. However, if both the neural proposal and the reference sampler fail to visit a cryptic basin (e.g. due to shared kinetic trapping across unconverged chains), both forward and reverse gaps remain blind. Evaluating G_{rev} across independent reference chains certified by convergence diagnostics (such as $\hat{R} < 1.05$) helps ensure that observed gaps exceed Monte Carlo noise.

Compute budgets and hardware normalization. The spin-glass benchmark equalizes total energy-evaluation budgets across methods (10^5 calls), incorporating neural training iterations into the cost. Wall-clock measurements reflect single-thread execution for reproducible comparison, but do not capture batched GPU parallelism during production inference where amortized neural samplers gain substantial throughput.

9 Conclusion

Self-normalized diagnostics such as the effective sample size and delta-method confidence intervals certify that a proposal fits the regions it visits, and nothing else (Proposition 1). When a generative sampler omits an entropic or energetic basin carrying Boltzmann mass α, the ESS remains high and nominal intervals shrink around a biased estimate, collapsing empirical coverage toward zero.

Against exact combinatorial ground truth on spin glasses, masked diffusion and GFlowNets exhibit severe calibration collapse in the frustrated phase (covering the true free energy in only 8% and 47% of runs at $\beta = 2$), while VAN-MADE and the classical interval estimators (AIS, SMC, PA) stay near 95%. A flow on an analytic alanine surrogate drops the $C_{7\mathrm{ax}}$ basin (8.1% of the surrogate's mass) while reporting an ESS of 79% and coverage of 10%, and flows on model landscapes in up to 414 dimensions behave the same way. The reverse check flags these omissions, but a control with the correct support can also give a large reverse gap.

In molecular units the shift depends on the dropped basin's mass: omitting the Trp-cage native basin, 50.3% of our trajectory, would shift F by 0.417 kcal/mol, while a basin of a few percent shifts it by a few hundredths. Unbiased MD trajectories share the blind spot: they report nothing about funnels they fail to reach.

Our findings suggest that self-reported forward metrics should be supplemented with independent bidirectional checks, particularly when evaluating samplers on complex physical systems where ground truth is unknown, and that new architectures should be tested against solvable references where mode drop can be proven.

Recommended evaluation protocol for practitioners. To prevent false confidence in scientific applications, we recommend a four-point evaluation protocol: (1) *Do not rely on forward metrics alone*: high

ESS certifies only local fit within visited basins, providing zero signal of omitted Boltzmann mass. (2) *Deploy bidirectional checks*: evaluate the reverse gap G_{rev} against an offline library of independent reference configurations (with mixing verified via $\hat{R} < 1.05$) before high-throughput deployment. (3) *Pre-flight audit on solvable exact references*: benchmark new neural architectures on frustrated systems with exact partition functions (such as SpinGlassBench or low-dimensional surrogates solved by quadrature) to expose mode truncation. (4) *Check populations against converged data*: compare basin fractions with enhanced-sampling or long equilibrium simulations of the same system and force field, not with single trajectories. Crucially, self-normalized diagnostics should never be treated as proof of convergence without orthogonal validation.

Reproducibility

`scripts/make_figures.py` generates every figure, Tables 1 and 2, and every data-derived number in the text (as macros in `paper/generated/values.tex`) from the files in `data/`: the spin-glass summary (`summary_sk24.json`, from the companion SpinGlassBench runs), per-frame RMSD and radius of gyration of the protein trajectories, the LJ_{38} sweep, and the stored results of the flow experiments. The toy double well is computed in that script with fixed seeds. The 2D alanine flow is produced by `scripts/eval_alanine_flow.py` (CPU, minutes); the flows on the higher-dimensional model landscapes (alanine surrogate, 66D; two-basin LJ_{38} on the real potential, 114D, RealNVP and $SE(3)$-invariant; two-Gaussian 408D model) by `scripts/eval_3d_molecular_flows.py` and `scripts/eval_3d_equivariant_and_khat.py` (GPU). Not in this repository: the raw MD trajectories, the OpenMM simulation scripts, and the spin-glass benchmark runs, which are in the companion SpinGlassBench repository. The literature populations quoted in Section 7 are typed in and unverified. In this version the Pareto $\hat{k}$ values that were not computed from the flows' own weights are withdrawn; the stored file records the withdrawn values and why. Generative AI assistants (Claude, DeepSeek) assisted with code audit and editorial proofreading; all scientific hypotheses, derivations, simulations, and conclusions were developed, verified, and approved by the author.

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